

AMENDMENT NO. 1 SEPTEMBER 2020

TO

IS 9473 : 2002 RESPIRATORY PROTECTIVE DEVICES — FILTERING HALF MASKS TO PROTECT AGAINST PARTICLES — SPECIFICATION

(*First Revision*)

(*Page 1, Clause 2*) — Insert the following *after* the last row:

17274 (Part 3) : 2019/	Respiratory protective devices — Methods of test and
ISO 16900-3 : 2012	test equipment: Part 3 Determination of particle filter penetration

(*Page 2, Clause 5.4.2, Table 1, NOTE*) — Substitute the following *for* the existing:

NOTE — A total of 9 filtering half masks shall be tested for each aerosol.

- a) Testing in accordance with **A-4** using the short-term penetration test in IS 17274 (Part 3) shall be performed on:
 - 1) 3 samples as received, and
 - 2) 3 samples after simulated wear treatment as per **A-1.1**.
- b) Testing in accordance with **A-4** using the full exposure penetration test in IS 17274 (Part 3) shall be performed on:
 - 1) 3 samples after test for mechanical strength as per **A-10**, followed by temperature conditioning as per **A-1.2**.

(*Page 3, Clause 5.11, Table 2, Last Row*) — Substitute the following *for* the existing:

1 bar = 10⁵ Pa; 1 mbar = 100 Pa

(*Page 4, Clause 7.6*) — Substitute the following *for* the existing:

The instructions shall indicate that single-use filtering half masks shall be discarded after single-use. For masks intended for more than single-use, manufacturers shall

provide instructions/indication for discarding mask after a specified period/time of use. The manufacturer shall also provide detailed procedure for disposal of masks as per Solid Waste Management Rules, 2016 notified by Ministry of Environment, Forest and Climate Change.

(Page 5, Clause 8.3, Table 4) — Substitute the following for the existing:

Table 4 Summary of Tests

Sl No. (1)	Requirement (2)	No. of Samples ^a (4)	Conditioning ^b (5)	Test Clause (6)
i)	Visual inspection	All	AR	A-11
ii)	Material	6	SW (3) TC (3)	A-11
iii)	Cleaning and disinfection	5	AR	Manufacturer's instructions
iv)	Practical performance	2	AR	A-2
v)	Total inward leakage	10	AR (5) TC (5)	A-3^c
vi)	Filter penetration	9 (for each aerosol)	AR (3) SW (3) MS + TC (3)	A-4^c
vii)	Flammability	4	AR (2) TC (2)	A-5
viii)	CO ₂ content of inhalation air	3	AR	A-6^c
ix)	Field of vision	2	AR	A-2 & A-8 of IS 14746
x)	Exhalation valve flow	3	AR (1) TC (2)	5.10.2.3 (FC)
xi)	Exhalation valve pull	3	AR (1) MS (1) TC (1)	A-7
xii)	Breathing resistance (valved)	12	AR (3) SW (3) TC (3) FC (3)	A-8^c

Table 4 (Concluded)

SI No. (1)	Requirement (2)	No. of Samples ^a (4)	Conditioning ^b (5)	Test Clause (6)
xiii)	Breathing resistance (valveless)	9	AR (3) SW (3) TC (3)	A-8^c
xiv)	Clogging (Optional for single-use masks)	3	AR (1) TC (2)	A-9 A-8^c A-4^c

^a Most samples are used for more than one test

^b Conditioning abbreviations:
AR – As received
SW – Simulated wearing treatment
TC – Temperature conditioned
FC – Flow conditioned

^c Wherever applicable, equivalent automated instruments validated against the corresponding test methods prescribed in Annex A may be used. For total inward leakage, even if an automated instrument is used, the test subject shall follow the exercises prescribed in **A-3**.

(Page 11, Clause A-4) — Substitute the following *for* the existing:

A-4 PENETRATION OF FILTER MATERIAL

The particle filtering half mask shall be mounted in a leak tight manner on a suitable holder for the test(s), ensuring that components of the mask that could affect filter penetration values, such as valves and harness attachment points are exposed to the challenge aerosol.

Testing of short-term penetration and full exposure penetration shall be done in accordance with IS 17274 (Part 3) at test aerosol flow rate of 95 l/min.

(Page 18, Clause **A-8.1**) — Substitute the following *for* the existing:

A-8.1 Test Samples and Fixture

A-8.1.1 Valveless Particle Filtering Half Mask

A total of 9 valveless particle filtering half masks shall be tested: 3 as received, 3 after temperature conditioning in accordance with **A-1.2**, and 3 after the simulated wearing treatment as described in **A-1.1**.

A-8.1.2 Valved Particle Filtering Half Mask

A total of 12 valved particle filtering half masks shall be tested: 3 as received, 3 after temperature conditioning as per **A-1.2**, 3 after simulated wearing treatment as per **A-1.1**, and 3 after flow conditioning as per **5.10.2.3**.

The particle filtering half mask shall be fitted securely in a leak tight manner but without deformation to the dummy head (type Sheffield).

(Page 20, Clause **A-8.2**) — Insert the following paragraph *after* the 1st paragraph:

The exhalation resistance is measured with the dummy head placed successively in the following 5 defined positions:

- a) Facing directly ahead;
- b) Facing vertically upwards;
- c) Facing vertically downwards;
- d) Lying on the left side; and
- e) Lying on the right side.